



MCAST

Machine Learning and Statistical Techniques in Investment Decision Support

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in Software Development**

Authorship Statement

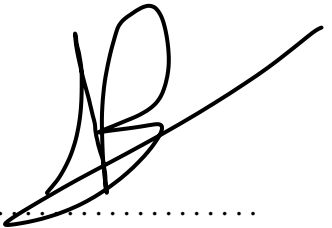
This dissertation is based on the results of research carried out by myself, is my own composition, and has not been previously presented for any other certified or uncertified qualification.

The research was carried out under the supervision of Mr. Alvin Cardona.

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Abstract

Investors constantly strive to predict future market trends in order to maximize their profits. However, the stock market's highly volatile nature poses a significant challenge when it comes to accurately forecasting stock fluctuations. As a result, successful investment strategies require a considerable investment of time and effort in gathering and analyzing relevant information about specific stocks to make well-informed decisions. This has generated significant interest within the academic community regarding the use of machine learning methods for stock market prediction.

While previous research in this field has shown promising results, it has yet to meet the required standards for developing investor support tools that can make optimal investment decisions on behalf of investors. The proposed technology addresses this by combining machine learning and natural language processing techniques in a complementary manner. It incorporates an optimized data storage engine that employs data serialization, memory caching, and index utilization to achieve efficient query processing. Additionally, multiple crawling components are utilized to extract freely available data from Yahoo Finance, including news articles, stock price history, and stock details. Moreover, the technology employs a hybrid machine learning model that integrates weighting concepts from artificial neural networks and clustering concepts from K-means classifiers. This approach enhances the accuracy and effectiveness of the predictions.

In this research, a proof of concept tool was developed to evaluate the pro-

posed technology; specifically in predicting the 30-day outcome of 221 stocks from the XETRA stock exchange. This study outperformed previous research endeavors that had achieved MSE error margins ranging from 0.00109 to 0.03526, by achieving MSE margins of 0.00001099 to 0.02880356. Hence, this research signifies a significant improvement and surpasses the earlier findings. The accomplishment of these results was made possible by utilizing 3 years of training data, which enabled a thorough analysis and enhanced the accuracy of the predictions. These collective findings not only confirm the effectiveness of the proposed technology but also establish a robust foundation for future explorations and advancements in this particular field.

Keywords: Machine Learning, Artificial Neural Networks, K-Means Classifiers, Stock Market, Natural Language Processing, Financial News Analysis, Web Crawlers, Stock Price Predictions.

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Acronyms

NLP	Natural Language Processing
NER	Named Entity Recognition
PIP	Perceptually Important Points
GA	Genetic Algorithms
ANN	Artificial Neural Networks
LSTM	Long Short-term Memory
RST	Rough Set Theory
MSE	Mean Squared Error
MAPE	Mean Absolute Percentage Error
VADER	Valence Aware Dictionary and Sentiment Reasoner
ML	Machine Learning
SVM	Support Vector Machines
IDE	Integrated Development Environment
OOP	Object Oriented Programming
RSS	Really Simple Syndication

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Chapter 1

Introduction

1.1 Scope of Research

The stock market has a significant influence on both individual investors' interests, as well as a country's entire economy trend [1]. As a result, the prediction of stock movement plays a vital role within the stock trading sphere. Investors continuously attempt to anticipate future market trends in order to maximize profits [2]. However, the prediction of stock fluctuations is widely recognized as a formidable task due to the stock market's highly volatile nature [1] [3]. Several key factors may affect the stock price of a particular company, such as financial statements, breaking news stories, sector performance, market sentiment, etc. [4]. Therefore, successful investment strategies require a substantial investment of time and effort in order to gather and analyze relevant information regarding specific stocks, in order to make informed decisions [5]. Technical analysis and Fundamental analysis are two of the main methodologies that financial practitioners employ when making trading decisions and forecasting stock prices, some combining the both to obtain increased benefit [6].

Advancement in machine learning technology in the field of stock market prediction has provided investors with trading support tools to assist them in their stock trading ventures [7]. This is particularly useful for retail investors who are

often limited by time and resources. This highlights the importance of having dependable investment support technology in today's investment landscape. The domain of stock market prediction based on machine learning methods has therefore garnered significant attention from academic communities [6]. However, despite yielding relatively encouraging results, prior research in this field [8] [9] [1] [10] [11] has not yet met the required standard for investor support tools to make optimal investment decisions on behalf of investors [4].

Hence, this research aims to establish, and potentially enhance, the current state of investment technology, with a specific focus on analysis techniques that can work with freely available data. In pursuit of this goal, the objective is to propose appropriate techniques that can provide practical assistance to retail investors. Further, the researcher aims to leverage the advancements in machine learning technologies by integrating them into the analysis process, thereby enabling retail investors to make better-informed investment decisions. Hence, this study seeks to improve the efficacy of investment analysis, ultimately benefiting retail investors in their pursuit of optimal investment strategies.

1.2 Motivation

The researcher is driven by a personal interest to explore and contribute to the field of stock market predictions. Fascinated by the complex dynamics of financial markets, the researcher is determined to employ machine learning algorithms to enhance the efficacy of stock market predictions. By undertaking this study, the researcher aims to bridge the gap between machine learning techniques and its application to stock market analysis, thereby contributing to the field and providing valuable insights for both academic and practical purposes.

1.3 Aims and Objectives

The primary aims of this research could be summarised as follows:

- Establish the current state of art of machine learning and associated technologies in stock trading.
- Evaluate its performance by examining the relationships between stocks, time, and events.
- Investigate the feasibility of enhancing this technology, from the perspective of retail investors.

1.4 Research Questions

This study sought to answer these following research questions:

- What level of accuracy is achievable when machine learning is applied to technical and fundamental strategies in predicting future stock prices?
- Does custom NLP-ML integrated technology perform better than off the shelf components in stock price prediction?

1.5 Ethical Considerations

This research proposes appropriate techniques that can provide practical assistance to retail investors, allowing them to make their own investment decisions. The research is software-related, eliminating any risk of physical harm, and there will

be no target participants or human interaction involved. Instead, data will be collected from freely available online sources like websites, discussion forums, and social media platforms. This ensures that no harm will be caused to businesses, and moreover, no confidential data will be divulged.

1.6 Research Outline

This section provides an overview of the remaining chapters, consisting of four main sections that will guide the structure and organization of the study. Chapter 2 (Literature Review) provides a synthesis of past research contributions to the field of stock market prediction. It explores existing knowledge and identifies gaps or limitations in the literature. Chapter 3 (Research Methodology) presents a detailed explanation of the methodological framework adopted for evaluating the proposed technologies. This section outlines the data collection and analysis methods employed, providing a rationale for their selection. Chapter 4 (Analysis of Results and Discussion) presents the findings of the study and engages in a comprehensive discussion, relating the results to the research objectives and comparing them with the existing literature. Finally, chapter 5 (Conclusions and Recommendations) summarizes the dissertation, emphasizing the key findings, limitations, and strengths of the research. Additionally, this chapter offers recommendations for future research directions, contributing to the advancement of knowledge in the field of stock market prediction.

Chapter 2

Literature Review

2.1 Overview

This study aims to evaluate the performance of machine learning when applied to technical and fundamental strategies in price movement prediction. Hence, literature regarding the following topics was reviewed: (a) types of investment products, (b) forms of quantitative and qualitative techniques used to anticipate market trends and (c) the machine learning algorithms utilized in aiding trading decisions.

2.2 Stock Market

The stock market is a network of stock exchanges where investors exchange stocks; assets that represent fractional ownership in a company. Like any other economic market, the stock market employs fundamental economic principles such as supply and demand, short selling, trading time frames, charting techniques, and so on [5]. Trading stocks entails purchasing shares of a company and selling them on a stock exchange to make a profit. This exchange benefits both corporations and investors, as the company's value is directly proportional to value of a stockholder's shares [5]. A stock market index is a subset of the stock mar-

ket, used to gauge the performance of a country, region, industry or category of stocks. Therefore indices are influential within the stock market as they measure stocks collectively and thus track how the overall sub-market is performing [12].

2.3 Foreign Exchange

Foreign exchange (Forex) is the term used to describe the conversion of one currency to another, and it represents one of the largest financial markets in the world. Hence, traders attempt to anticipate exchange movement trends in order to maximize returns and minimize risks [13]. However, similarly to stocks, the exchange rate is influenced by a variety of factors, including a country's economy, politics, inter-market relationships, and so forth, thus making the forecasting of Forex trends difficult [14] [13].

2.4 Cryptocurrency

Cryptocurrencies are digital currencies traded among individuals or groups, using network-based technology that employs cryptographic algorithms to secure transactions [15]. They possess characteristics of decentralization, transparency, and immutability due to their reliance on blockchain technology [15]. Their prices are highly volatile and influenced by factors such as transaction costs, mining difficulty, market trends, popularity, alternative coin prices, stock markets, sentiments, and legal considerations [16]. These factors contribute to rapid fluctuations in cryptocurrency prices, posing challenges for researchers attempting to forecast them [16].

2.5 Trading and Investment Tools

Existing research shows that machine learning methods can make use of a company's stock data to extract key trends and tentatively predict the direction of the financial market [17]. These implementations utilized both technical analysis (which examines past stock prices, volumes and other parameters in order to identify trends) and fundamental analysis (which examines the performance and state of the company behind the stock to predict how the company will perform in the future). Both are necessary to anticipate future market trends and price accordingly [18].

2.5.1 Technical Analysis

Technical analysis is a technique that involves looking at stock prices' past performance, trading patterns and other related elements. This analysis is based on the principle that history repeats itself, and it can be used to predict market direction in the future [19]. Such forecasts could make use of data such as price, volume and interest rates. While these mechanisms cannot predict the exact price at which a stock will trade at any given time, they can identify trends and provide insight into whether it will rise or fall and at what rate [20]. However, Technical Analysis only considers quantifiable information and neglects several factors such as various forms of news, conferences and discussions, all of which can influence investors' attitudes and sentiment toward the market as a whole, or specific stocks. These unrelated factors are difficult to predict, but they have the potential to influence market trends and thus stock prices [21].

2.5.2 Fundamental Analysis

Fundamental analysis is a technique in which the factors which affect the stock price, such as the performance of an organization and the profitability generated, are thoroughly examined. Fundamental analysis factors include data from the economy (inflation, unemployment, debt level, etc.), the industry sector (including competitors and partners/suppliers) and the company itself (earnings, management effectiveness, returns on equity) [10] [22]. By evaluating these influential factors, the relative value of a company's shares could be estimated. The stock price valuation can be used to determine whether the stock is at an acceptable price and is not over or undervalued. Furthermore, by understanding an organization's share price, one can predict the expected returns on investment and make a more informed investment decision [20].

2.6 Stock Market Prediction Techniques

Previous studies in this field primarily focused on the prediction of stock price movement (identifies whether prices exhibit a bullish or bearish trend), rather than attempting to predict the actual price [2]. Despite several scientific attempts, no method has attained a level of accuracy in stock price prediction that could serve as a substitute for human reasoning. However, despite the lack of consistent predictions, there have been some mild successes [4]. Stock market research encapsulates the two trading philosophies mentioned above; fundamental and technical approaches. Technicians utilize charts and modeling techniques in order to identify patterns in price and volume. These individuals rely solely on historical data to forecast future outcomes [19].

However, one area of limited success comes from qualitative information. Data extracted from quarterly reports, news stories or online forums can drastically af-

fect the share price of a stock [4]. The majority of existing financial text mining (the process of extracting valuable insights from textual data) literature relies on the use of Natural Language Processing (NLP). NLP is a key tool in sentiment analysis as it enables machines to interpret human language [23]. Named Entity Recognition (NER) is a sub-field of NLP that involves identifying, extracting and classifying named entities from text. Hence, using this technology, researchers are able to extract meaningful insights from large amounts of unstructured qualitative data [24] [23]. These techniques are typically paired with machine learning algorithms, whereas weights that are proportional to the movement of a stock's price, are assigned to a predefined set of keywords. These forms of analyses have shown a clear but restricted capability to predict the direction of stock prices [4].

2.6.1 Pattern Recognition

Pattern recognition is a technical analysis technique that involves the use of algorithms to identify recurring stock market patterns from stock price history [3]. Figure 2.1 shows some formations that emerge in the price movement of a financial instrument over time. Studies have found some correlation between patterns

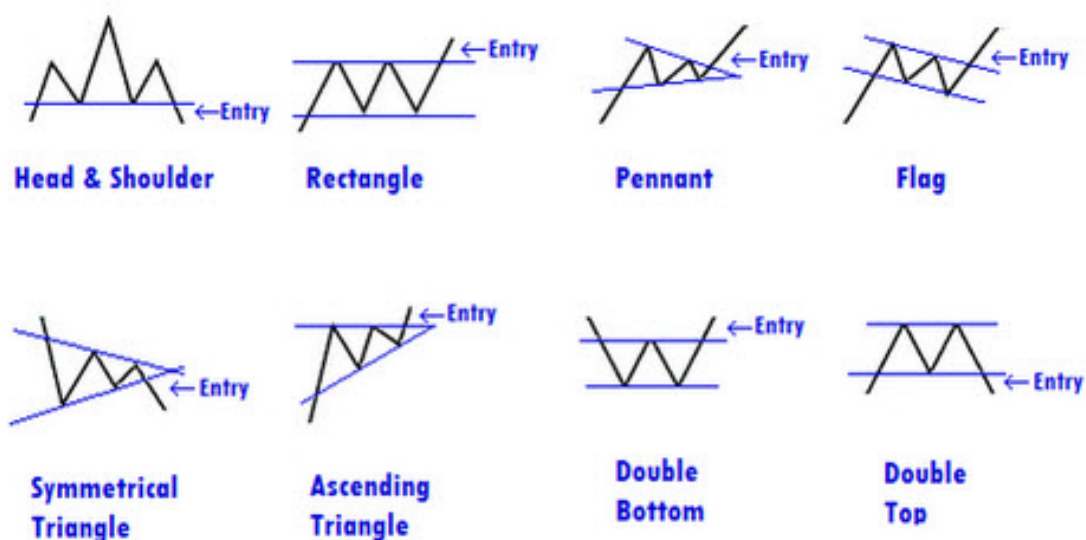


Figure 2.1: Types of Stock Chart Patterns

and future trends. One study by Chen and Chen [9] suggested a hybrid method that can be used to predict future market movements by identifying bullish (up-trend) flag patterns on the NASDAQ index. The researchers [9] created a methodology that combines two established pattern recognition techniques, perceptually important points (PIP) and template matching. The PIP technique is characterized by the reduction of time-series dimensions by preserving the salient points, whilst template matching is a technique used to match a given stock pattern with a reference image [9]. Their suggested hybrid strategy performed significantly better than other models like Genetic Algorithms (GA), Rough Set Theory (RST), and a hybrid of GA and RST [9].

Previous studies by [25] and [26] opted to employ an Artificial Neural Network (ANN) model to detect stock patterns, describing the template-matching approach as constraining due to the required participation of domain experts during the implementation process. Zhang and Lou [8] proposed a system that applies an ANN model to the classification and prediction of stock price patterns. The ANN model returned satisfactory results with a 73.29% prediction accuracy, outperforming a fuzzy algorithm which returned 62.12%. Moreover, [8] have expressed their intention to further develop their current implementation, taking into consideration external factors that may have a significant impact on the stock price.

Overall, the literature suggests that pattern recognition is an effective tool for predicting stock market trends. The ANN deep learning model, similar to the template matching and PIP pattern recognition techniques, have shown to be particularly effective in this area [8] [9].

2.6.2 Financial News Analysis

It has been established that a correlation exists between the publication of financial news and stock price fluctuations. Hence, research works exist on the analy-

sis of financial news articles and other textual sources to extract insights, which are then used to predict future price movements, with the use of NLP [27] [28].

Several studies have investigated the effectiveness of this fundamental analysis approach, with promising results. For instance, a study by Guo [1] proposed an Long short-term memory (LSTM) neural network model with news sentiment scores and historical stock data as input, to predict future stock close price. This research utilized NLP tools to quantify the effect from news articles by the New York Times, thus translating them into quantitative weights. This research utilized Valence Aware Dictionary and Sentiment Reasoner (VADER), a subset of NLP technology designed to analyze the polarity of short texts, and TextBlob, a Python library used to perform NLP tasks [1]. VADER is based on a lexicon that measures the effect of text by matching sentence features to sentiment scores. Since it was designed to evaluate short sentences, VADER was used to analyze article headlines, while TextBlob was used to analyse news article content. Ultimately, the model returned the mean squared error (MSE) values of 0.00109, 0.00497 and 0.03526 (the closer the MSE value is to 0, the better the model's performance) when attempting to predict the closing stock price of Amazon, Microsoft and Tesla respectively [1].

Bharti et al. [29] employed a similar approach to Guo [1] in their study. They utilized a data crawler to collect a rich dataset consisting of news headlines and the corresponding market prices, thus enabling a holistic examination of the relationship between news sentiments and stock market behavior. Sentiment analysis of the news headlines was also conducted using VADER [29]. Following this, the sentiment-enhanced dataset was utilized in two separate prediction models. Firstly, XGBoost was employed for classification purposes to anticipate stock price movements (bullish or bearish). Secondly, Long Short-Term Memory (LSTM) was used for more accurate stock price predictions [29]. In terms of performance, LSTM exhibited a Mean Absolute Percentage Error (MAPE) of 4.164%, while XGBoost achieved an accuracy score of 0.945. Accuracy score quantifies the model's cor-

rectness in predicting outcomes in relation to the total number of predictions made [29].

Other studies by Shah et al. [10] and Dang et al. [11] both analyzed the effects of news on the stock market with use of a dictionary-based sentiment analysis model. The use of a sentiment analysis dictionary allowed the researchers to achieve a directional accuracy of 70.59% and 73% respectively, in short-term stock price movement prediction [10] [11]. In addition to this, Shah et al. [10] mentioned the implementation of a word-weighting approach in the analysis pipeline as part of their future research. This technique assigns the appropriate weight to each term based on their impact to the stock price. Overall, the findings [10] [11] [1] suggest that news analysis can be an effective approach for predicting stock prices and stock price movement.

2.6.3 Social Media Sentiment Analysis

The evolution of social media has created a new meaningful channel for users to share information and their personal feelings. Opinions posted on social media platforms depict the sentimental state of several users [30]. These users express their opinions for a company or its products in tweets or comments. The extraction of sentiments from these comments is used to identify the user's view for a particular product or company [31]. Hence, machine learning researchers exploited this wealth of information.

In a study by Liu et al. [32], the researchers measured the synergy between investor sentiment and stock price movement. NLP techniques were utilized to extract opinions from China's largest social media platform for stock market investors. The researchers [32] found that investor sentiment has a weak correlation with stock prices. These findings are consistent with those of Behrendt and Schmidt [33] who used Twitter sentiment to measure investors' interest in

individual-level stocks. They claimed that the incorporation of information extracted from Twitter into intraday prediction models for return volatility is unlikely to have a substantial impact on prediction performance [33].

Moreover, Dhaoui et al. [30] identified numerous limitations in their evaluation of the performance of two automated social media sentiment analysis approaches (lexicon-based and machine learning). Such limitations include informal language, contextual understanding, misspelling, ironic language and sarcasm. Hence, while NLP has made significant progress, accurate analysis of social media content remains a formidable task due to the language used on these platforms [30].

2.7 Machine Learning Algorithms

Machine learning encompasses various algorithms that enable machines to identify patterns autonomously, thus eliminating the need for it to be programmed with specific instructions [34]. Machine learning (ML) has become an increasingly popular technique for predicting stock movements. This approach uses historical stock data to train models that can identify patterns and trends that may indicate future movements in the stock market [35]. The accuracy of these predictions can vary depending on the amount and quality of the data used to train the models, as well as the complexity of the algorithms used [36].

2.7.1 Artificial Neural Networks

Artificial Neural Networks (ANNs) are designed to replicate the neural networks in the human brain [34] [37] [38]. ANNs consist of interconnected neurons and are composed of multiple layers of nodes, including input and output layers specifically designed for data input and result collection, respectively [38].

The signals between the interconnected nodes carry weights, which are adjusted during the training process to enhance the neural network's performance [37].

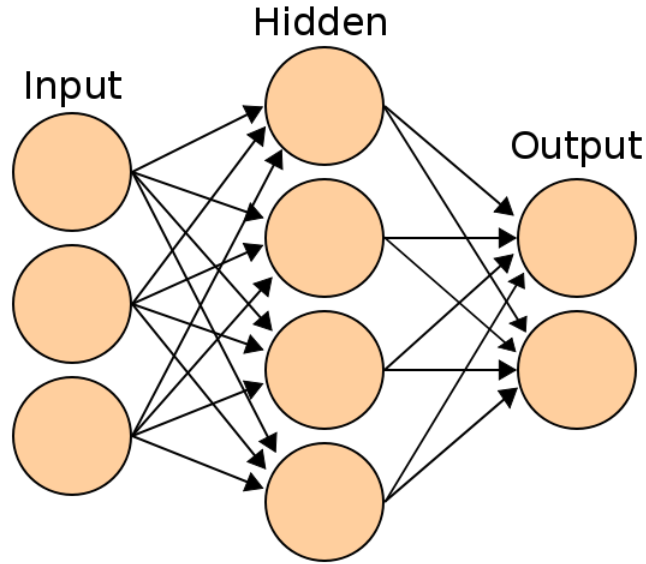


Figure 2.2: Artificial Neural Network

The ANN passes input data through the network's nodes, which rely on the incoming signal's weights in order to make predictions. The network is trained by utilizing training data and random weight parameters to enable the network to generate provisional output values [38]. Once the signals are fed through the input layer, they are transmitted through the hidden nodes, where their weights are adjusted, in order to determine the output values [37]. Subsequently, the output values are compared to correct values, and a self-learning process referred to as back-propagation is employed to adjust the weights accordingly, if the outputs are not within a predetermined threshold. This process is repeated until the network's generated outputs meet the specified threshold [37]. Conversely, excessive training can result in a network that is unable to generalize. This could potentially pose a problem in the application of stock price prediction [39]. Moreover, Yoo et al. [39] stated that ANNs outperform conventional statistical methods when dealing with incomplete data sets in large quantities. This is attributed by ANNs adaptive learning abilities that make them increasingly useful in the field of fi-

nancial predictions [39].

2.7.2 Support Vector Machines

Support Vector Machine (SVM) is a machine learning algorithm, used for classification and regression tasks, that aims to find a hyperplane in a higher-dimensional space to separate classes [40]. Huang et al. [41] conducted a study on financial movement prediction using SVM to forecast the weekly direction of the NIKKEI 225 index. In this study, the researchers found that SVM outperformed other classification methods such as Elman Backpropagation Neural Networks, Quadratic Discriminant Analysis and Linear Discriminant Analysis [41] [40]. Similarly, Kim [42] utilized SVM to predict the daily stock price change direction in the Korea stock price index (KOSPI). The results demonstrated that SVM surpassed both Backpropagation Neural Network and Case-Based Reasoning, highlighting its potential as an alternative approach to improve stock market prediction accuracy [42].

2.7.3 K-means

The K-means algorithm is an unsupervised clustering technique that aims to classify unlabeled data into distinct groups, referred to as clusters [43]. Objects are grouped into clusters based on their similarity, while dissimilar objects are assigned to separate clusters. This process ensures the maximization of intra-class similarity while minimizing inter-class similarity [44]. The partitioning process starts by randomly assigning data objects to different partitions. Each iteration sees objects transferred to another partition, with a centroid, which is the average of all data objects within it, representing each partition. Each partition must have at least one data object, and every object is assigned to exactly one cluster [43].

K-means is widely used due to its ability to effectively handle various types of data [43] [44]. The algorithm starts by selecting k classes with k reference points as initial estimates. Then the data is partitioned into k distinct regions based on their respective distances from the reference points [44]. In addition, each region is assessed using the reference point as its centroid, and the data points are clustered based on their nearest neighbour. The algorithm then employs iterative adjustments of the initial reference points, minimizing the distance between them and their respective members [44].

2.7.4 Hybrid Models

According to Wu et al. [45], not all machine learning technologies are equally effective for specific tasks. Hence, to overcome the limitations of single-method algorithms, hybrid machine learning techniques (which combine different mechanisms) have been adopted [45]. In Bahrammirzaee's [46] study, it was indicated that the combination of multiple machine learning techniques into hybrid models has led to improved performance, when compared to single-method techniques, in the domain of financial prediction. In stock prediction systems, hybrid techniques can utilize both simple and complex elements, thus guaranteeing that only the most relevant features are considered, leading to improved efficiency and accuracy of the hybrid system [47].

The following table 2.1 summarises the achievements of previous researches:

Research	Algorithm	Technique	Data	Prediction	Accuracy	Metric
Chen and Chen, 2016	Hybrid (PIP bull-flag pattern matching and bull-flag template)	Pattern Recognition	Stock Price History	Stock Movement	454.29%	TIR
Zhou and Hu, 2009	ANN and PIP	Pattern Recognition	Stock Price History	-	-	-
Guo et al., 2007	ANN	Pattern Recognition	Stock Price History	-	-	-
Zhang and Lou, 2021	ANN	Pattern Recognition	Stock Price History	Stock Movement	73.29%	Percentage Accuracy
Guo, 2020	LSTM	NLP (VADER)	News	Stock Prices	0.00109 (Amazon), 0.00497 (Microsoft) and 0.03526 (Tesla)	MSE
Bharti et al., 2022	LSTM	NLP (VADER)	News	Stock Prices	4.164%	MAPE
Bharti et al., 2022	XGBoost	NLP (VADER)	News	Stock Movement	0.945	Accuracy Score
Shah et al., 2018	Dictionary-Based Model	NLP (dictionary-based)	News	Stock Movement	70.59%	Percentage Accuracy
Dang and Duong, 2016	SVM	NLP (dictionary-based)	News	Stock Movement	73%	Percentage Accuracy
Liu et al., 2023	ANN	NLP	Social Media	Stock Movement	-	-
Behrendt and Schmidt, 2018	-	NLP	Social Media	-	-	-
Dhaoui et al., 2017	RTextTools	NLP (dictionary-based and machine learning)	Social Media	-	0.77 lexicon-based and 0.78 machine learning	F scores

Table 2.1: Summary of Literature

Chapter 3

Research Methodology

3.1 Introduction

This chapter outlines the quantitative methods used to attempt to answer the following research questions:

- What level of accuracy is achievable when machine learning is applied to technical and fundamental strategies in predicting future stock prices?
- Does custom NLP-ML integrated technology perform better than off the shelf components in stock price prediction?

Authors [1] [32] [8] have proposed various machine learning approaches that utilize financial data to identify patterns and tentatively predict the direction or even the price of the stock market. Therefore, several factors were considered when selecting the specific approach for this study. Being a form of technical analysis, pattern recognition only considers quantifiable data, thus overlooking several factors such as news, conferences and discussions, all of which can majorly influence investors' attitudes toward the market [3] [21]. Conversely, the efficacy of NLP technology when applied to the extraction of social media sentiment is hindered by the use of informal language, contextual understanding, misspelling,

ironic language and sarcasm [30]. In addition, [33] [32] claimed that investor sentiment is weakly correlated to stock fluctuation, thus unlikely to have a significant impact on the performance of prediction models.

On the contrary, It has been established that a correlation exists between the publication of financial news and stock price fluctuations [27] [28]. Researchers [10] [11] [1] have shown that the analysis of financial news content, using NLP, is an effective approach in the prediction of stock prices and movement. Hence, this research will contribute to the body of knowledge by investigating the potential benefits of integrating machine learning techniques with financial news analysis to predict stock prices, with the anticipation that similar technology can be applied to other investment products.

To address the research questions above, this research proposes the incorporation of machine learning and NLP techniques to predict the actual stock prices of specific stocks, by analysing publicly available news articles and stock prices from Yahoo Finance. It is anticipated that market reactions vary across different types of news [48]. Therefore, the proposed system would cater for each stock individually. Further research in this area could prove beneficial for the community, providing potential investors with a tool that delivers organized and centralized recommendations on their investments.

3.2 Concept

The proposed technology aims to automatically collect data from Yahoo Finance, namely news articles and stock price history, analyse it and deliver predictions regarding future stock prices. Given the dynamic nature of the stock market, it becomes increasingly pivotal that this process is executed within a reasonable timeframe, whilst ensuring continuous data updates. This process will be further

explained in the following sections.

3.2.1 Data Storage

Given the rapidly fluctuating stock market environment, predictive models for stock price predictions necessitate significant speed. However, the extensive volume of textual data required for this research poses a challenge to relational databases, which are specifically designed for structured data. Consequently, the proposed technology employs a data storage engine in lieu of a relational database, which was deemed unfeasible for the proposed technology. This data engine optimizes data storage and retrieval by efficiently serializing data to disk, while maintaining selected data cached in memory for fast access. Indexes are used to speed up sorting and searching where appropriate. This approach maximizes data utilization while minimizing disk access, thereby enhancing the performance of the system. This concept was inspired by Yu et al. [49], who proposed a high performance indexing architecture for real-time online audio searching systems. By serializing data to disk, the researchers [49] achieved enhanced query performance and reduced memory usage.

3.2.2 Data Collection

The acquisition and continuous updating of historical prices and related news data are essential for the efficacy of the proposed tool. Hence, the proposed prototype incorporates dedicated web crawling components to extract data from Yahoo Finance.

A web crawler is a tool that extracts and collects data from web pages by scanning through their content. It operates by parsing, searching, and potentially refor-

matting the information found on the page [50]. In the case of the web crawler employed by [50], the researchers utilized a Regular Expression (Regex) based pattern matching technique, to seek substrings from parsed web content.

Yahoo Finance, as suggested by [51] and [52], was chosen as the data gathering solution due to its abundance of real-time and historical data, reliable analysis, and extensive news coverage from reputable sources such as Bloomberg, Reuters and Wall Street Journal, amongst others. The data provided is categorised by stock, hence facilitating the importation and classification process. Moreover, Yahoo Finance provides several API's that may assist in accomplishing various tasks.

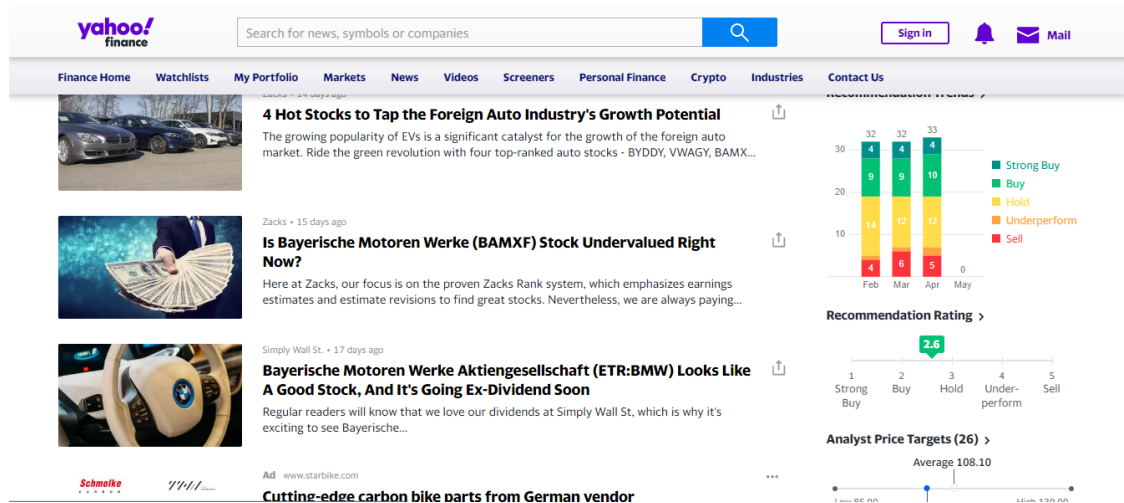


Figure 3.1: Yahoo Finance Web Design

3.2.3 Machine Learning

In Bahrammirzaee's study [46], it was indicated that the incorporation of multiple machine learning techniques into hybrid models resulted in enhanced performance within the field of financial prediction. This finding aligns with the statement made by Huang et al. [47], who emphasized the efficacy of hybrid techniques in stock prediction systems. Moreover, the researchers claimed that hybrid models

can effectively leverage both simple and complex elements, ensuring the consideration of only the most relevant features [47].

Similarly, in this study the researcher aims to integrate the most pertinent features from ANNs and K-Means, namely their respective weighting [37] and clustering [44] concepts, in conjunction with NER, for the purpose of quantifying the impact of news articles on a given stock's performance. The mechanism combines these elements by cross-matching the words extracted from news articles associated with a given stock, with the subsequent day's change in price. This process involves clustering similar words and assigning a weight to each word based on the change in price, determined by the number of data objects present within that cluster. A similar word weighting approach was suggested by Shah et al. [10] as part of their future work. Hence, by combining the strengths of ANNs and K-Means, this approach provides a comprehensive framework for the analysis of the influence of news articles on stock prices.

3.3 Proof of Concept Implementation

A proof of concept implementation has been developed to evaluate the proposed technologies. The tool was developed using the C# programming language, in the Visual Studio 2022 integrated development environment (IDE) as a Windows Forms Application. These choices were made to simplify the development by taking advantage of the IDE's Forms Designer while using the powerful constructs of the .NET framework. Throughout the development process, a 3-tier software development paradigm was followed, enabling a clear separation of concerns between the database, logic, and user interface components. This approach enhances modularity, allowing for easier scalability of the tool.

3.3.1 Pipeline

Figure 3.2 outlines the methodological principles that were involved in the research process. The technology being proposed starts from blank, and periodically and repeatedly crawls the Yahoo Finance RSS feed to retrieve news article links. These are then downloaded and processed, revealing names and tickers of stocks, which in turn are crawled from Yahoo Finance, in a recursive cycle. The machine learning process then looks for the impact of specific words on the stock prices, taking into account the popularity of the word, and the date when its article was published. This enables a machine learning model to be trained on the article content, respective stock, and price relationships. Thus, predictions can then be generated by using the model and applying the process in reverse on newly released news articles.

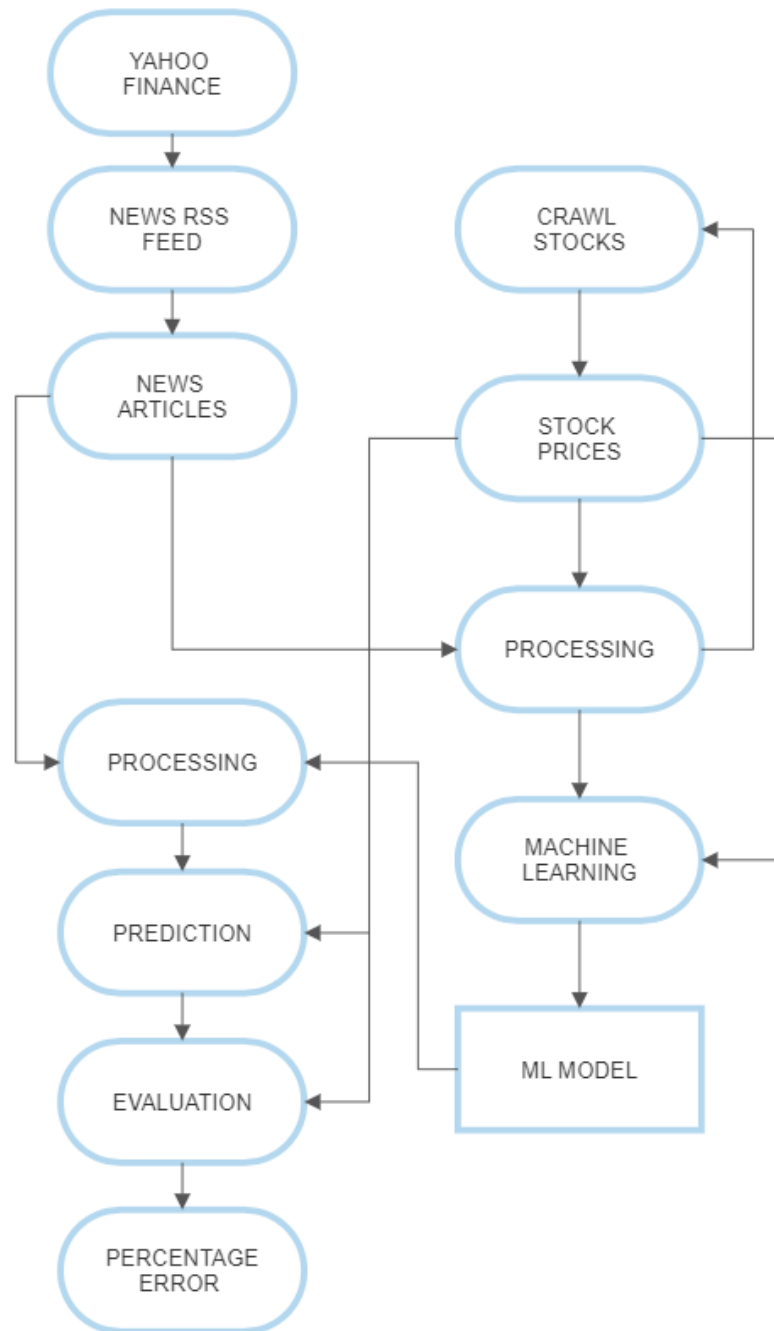


Figure 3.2: Overview of the Methodological Process

3.3.2 Data Engine

A custom data storage and processing engine was implemented as a substitute for a relational database in order to handle the substantial amount of textual data

required for this research at a reasonable speed. The main objective of this data engine is to optimize data storage and retrieval by efficiently serializing data to disk, while also maintaining a portion of the data cached in memory, for rapid access.

To facilitate the storage and manipulation of the data, three classes were designed to represent the three basic concepts associated with this type of data processing: *TimeSeries*, *Entity* and *EntityList*. Object Oriented Programming (OOP) concepts like inheritance and polymorphism are then used to extend these classes to create any data containers as needed. These superclasses are equipped with a wide range of data structure operations, enabling the system to add, update, and delete data effectively and efficiently. The *TimeSeries* class is specifically designed to store a sequence of data points collected over a given time interval, while the *Entity* class is utilized to store diverse data elements. The *EntityList* class allows the organised storage and retrieval of *Entity* instances, supporting any number of indexes as required. All these classes are initialized within the *Database* class, providing a centralized interface for accessing and managing the data. To improve the performance of sorting and searching operations, the implementation employs generic indexes that can be indexed by any field as required, such as timestamp, name, or ID.

Further, different filing modes were adopted based on the volume and type of data contained in each instance. The *SingleFileCached* mode was utilized for storing *Currencies* and *Stock Exchanges*. In this mode, a single file is employed for all the entities in the list, keeping a memory based cache of all of them, including indexes. This maximises the speed of access for entities that are used very frequently but are not too numerous, such as currency or stock exchange. Moreover, all the data related to each specific currency or stock exchange is cached in memory upon system startup, enabling efficient retrieval of data. For more numerous and dense entities such as *Stock*, the *MultiFileCached* mode was chosen. In this mode, a distinct file is created for each individual stock, thereby

allowing direct access to each entity's storage rather than having to update all the entities to save just one, like what happens with the SingleFileCached mode. Similar to the SingleFileCached mode, the data is cached in memory upon system startup for efficient retrieval, which again is appropriate for entities that are accessed very frequently, which is the case with Stock entities. However, entities that are very numerous and also contain a large amount of data such as the NewsArticle, are not feasible to keep cached, even more so considering that they are not accessed as frequently. For these types, MultiFile filing mode is used. Each entity is stored in a separate file, reflecting the significant volume of textual information.

To summarise, the implementation of this custom data storage engine addresses the research project's need to manage vast amounts of textual data. Through data serialization, caching of memory, utilization of indexes, and use of appropriate filing modes, the prototype achieves optimized data processing (storage, retrieval, and manipulation of data), hence improving the overall performance.

Figure 3.3 depicts the high-level structure of the main classes of the proof of concept implementation, of the proposed technology.

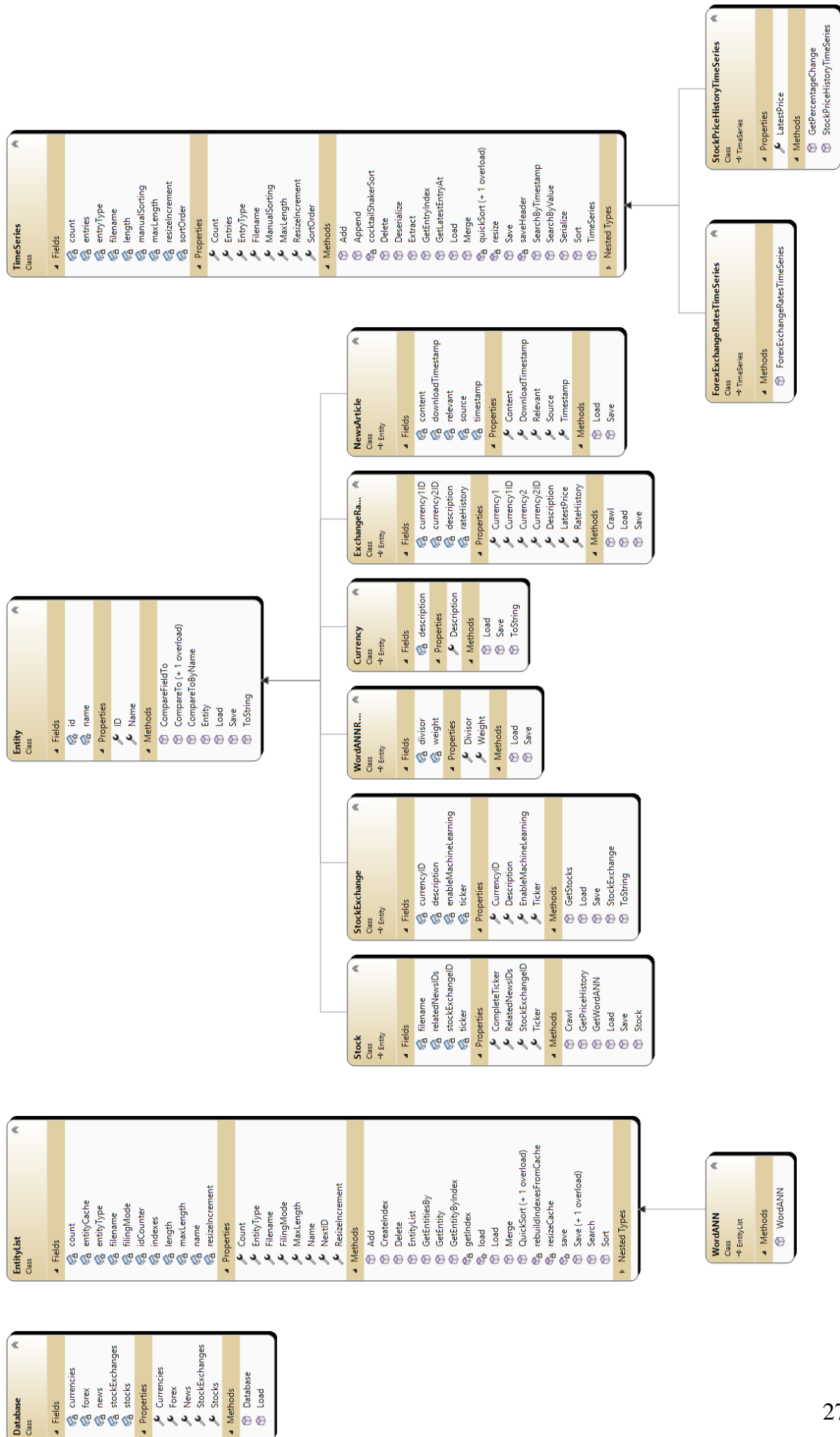


Figure 3.3: Class Diagram

3.3.3 Data Collection

For the purpose of this prototype, the researcher incorporated four distinct crawling components, each customized to extract a specific type of data obtained from Yahoo Finance. The subsequent sections provide a detailed examination of each crawling component.

Stock Price History Crawling

Stock price history is obtained by utilizing an API endpoint, provided by Yahoo Finance, which serves as a gateway to download a given stock's historical price data from their servers. By specifying the ticker symbol and unix date range parameters in the URL, a CSV file containing several fields depicting the specified stock's performance for the given period is retrieved and parsed. Using string manipulation techniques, the opening, highest, lowest, and closing prices, along with the trading volume for each trading day, are extracted and stored in a series of `StockPriceHistoryTimeSeries` instances, which are then stored in their respective `Stock` instances. This allows them to be updated and retrieved with other `Stock` data. As shown in Figure 3.4, the process can be invoked either to gather the price history of an individual stock or to automatically repeat it for all stocks currently saved in the system. By doing this, the tool's stock price history is consistently maintained and updated.

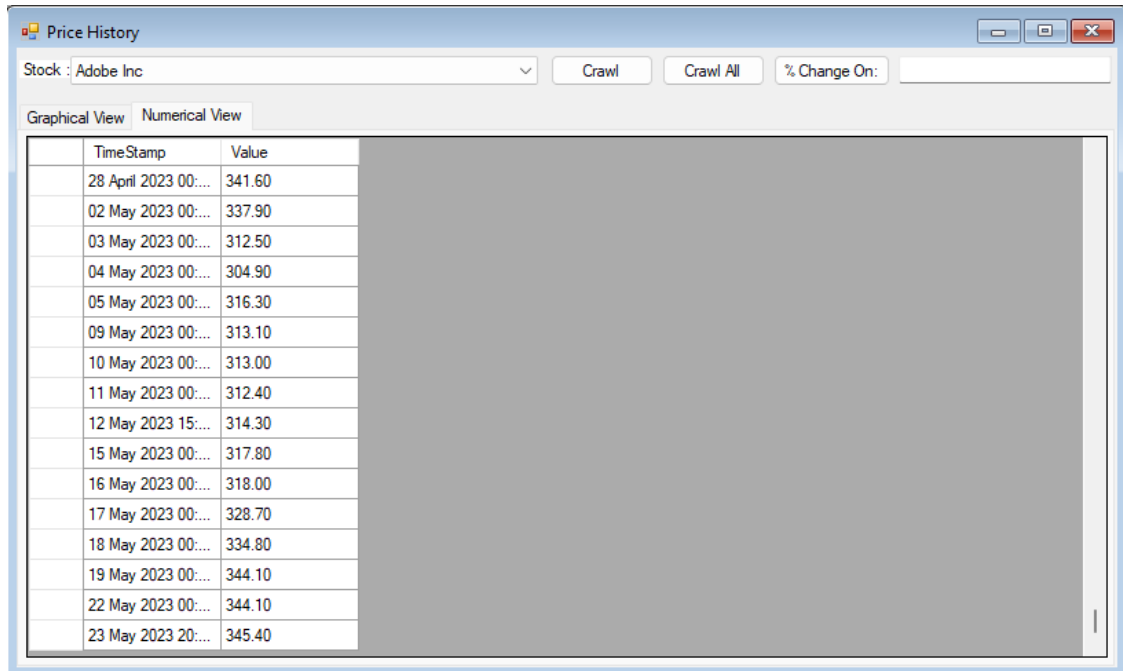


Figure 3.4: Stock Price History Form

Foreign Exchange Rate History Crawling

The design of the foreign exchange rate crawler resembles that of the stock price history crawler, utilizing the same API endpoint albeit with different query parameters. Instead of specifying a ticker symbol and date range, the foreign exchange rate crawler requires two three-letter currency codes (ex. USD for US Dollar, EUR for Euro) and the desired timeframe as input parameters. By utilizing the API endpoint, users have the ability to retrieve a CSV file that contains historical price data for a specific exchange rate and time period, that is then parsed. String manipulation techniques are employed to extract various data points such as opening, highest, lowest, and closing prices for each trading day. These fields are then stored in a series of `ForexExchangeRateTimeSeries` instances, which are then stored in their respective `ExchangeRate` instances, allowing them to be updated and retrieved with other `ExchangeRate` data.

	A	B	C	D	E	F	G
1	Date	Open	High	Low	Close	Adj Close	Volume
2	13/04/2023	105.84	107.49	105.84	107.43	107.43	24843600
3	14/04/2023	106.89	108.94	106.84	108.87	108.87	26578000
4	17/04/2023	104.66	106.16	104.52	105.97	105.97	37571200
5	18/04/2023	106.49	106.54	104.07	104.5	104.5	26596400
6	19/04/2023	103.58	104.98	103.07	104.18	104.18	20905700
7	20/04/2023	103.91	106.25	103.87	105.29	105.29	27820800
8	21/04/2023	105.47	106	104.78	105.41	105.41	25800100
9	24/04/2023	105.49	106.63	104.7	105.97	105.97	23542800
10	25/04/2023	105.83	106.69	103.84	103.85	103.85	46664100

Figure 3.5: Price Data Retrieved in CSV Format

Stock Crawling

The stock crawler retrieves stock data from a specified stock's webpage on Yahoo Finance. First, the ticker undergoes a series of validations to ensure it a) follows ticker criteria; b) is not an acronym in the context of the article and c) is not already stored in the database. If valid, the ticker is used as a parameter in the Yahoo Finance Search endpoint API, allowing the retrieval of JSON data pertaining to a list of equities with similar tickers, similar to the Yahoo Finance search bar. After converting the JSON data to a string, it undergoes string processing, following a standard pattern to extract all the ticker symbols present in the string.

```
{
  "explains": [],
  "count": 15,
  "quotes": [
    {
      "exchange": "NGM",
      "shortname": "Renalytix plc",
      "quoteType": "EQUITY",
      "symbol": "RNLX",
      "index": "quotes",
      "score": 20096.0,
      "typeDisp": "Equity",
      "longname": "Renalytix plc",
      "exchangeDisp": "Frankfurt",
      "sector": "Consumer Cyclical",
      "industry": "Auto Manufacturers",
      "isYahooFinance": true
    },
    {
      "exchange": "FRA",
      "shortname": "RENAULT INH.",
      "quoteType": "EQUITY",
      "symbol": "RNL.F",
      "index": "quotes",
      "score": 20096.0,
      "typeDisp": "Equity",
      "longname": "Renault SA",
      "exchangeDisp": "Frankfurt",
      "sector": "Consumer Cyclical",
      "industry": "Auto Manufacturers",
      "isYahooFinance": true
    },
    {
      "exchange": "PAR",
      "shortname": "RENAULT PFRN24OCT49",
      "quoteType": "EQUITY",
      "symbol": "RNL.PA",
      "index": "quotes",
      "score": 20095.0,
      "typeDisp": "Equity",
      "longname": "Renault SA",
      "exchangeDisp": "Paris",
      "sector": "Consumer Cyclical",
      "industry": "Auto Manufacturers",
      "isYahooFinance": true
    },
    {
      "exchange": "STU",
      "shortname": "Nordea 1 Nordic",
      "quoteType": "EQUITY",
      "symbol": "RNL.SG",
      "index": "quotes",
      "score": 20003.0,
      "typeDisp": "Fund",
      "longname": "Nordea 1, SICAV - Nordic Equity Fund",
      "exchangeDisp": "Stuttgart",
      "sector": "Consumer Cyclical",
      "industry": "Auto Manufacturers",
      "isYahooFinance": true
    },
    {
      "exchange": "STU",
      "shortname": "Nordea 1 Nordic",
      "quoteType": "EQUITY",
      "symbol": "RNL.SG",
      "index": "quotes",
      "score": 20002.0,
      "typeDisp": "Equity",
      "longname": "Renault SA",
      "exchangeDisp": "Stuttgart",
      "sector": "Consumer Cyclical",
      "industry": "Auto Manufacturers",
      "isYahooFinance": true
    }
  ]
}
```

Figure 3.6: Tickers in JSON Data

Next, any extracted tickers are crawled to gather relevant data pertaining to that stock. As illustrated in Figure 3.7, by formatting the Yahoo Finance URL with

a stock ticker parameter, the researcher has access to HTML code containing the stock exchange name, stock exchange ticker, currency and stock name that correspond to the ticker parameter.

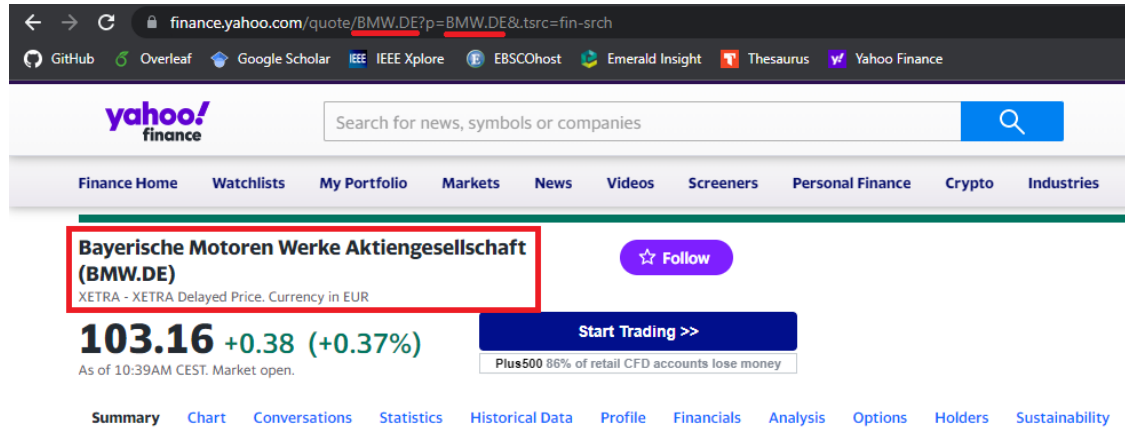


Figure 3.7: Stock Related Data

The HTML code is downloaded and parsed, from which specific patterns containing the stock's details can be identified and withdrawn. This routine is initiated during the news processing, which is further examined in the next section. Ultimately, the stock price history crawling component is invoked to gather the price history corresponding to the recently crawled stock.

```
</div><div class="C($tertiaryColor) Fz(12px)"><span>XETRA - XETRA Delayed Price. Currency in EUR</span></div>
lor):h " data-test="STATISTICS" role="none"><a class="Lh(44px) Ta(c) Bdbw(3px) Bdbb(s) Px(12px) C($linkColor)"
```

Figure 3.8: Use of Pattern Matching

News Crawling

A limitation imposed by Yahoo Finance's website design is that not all news articles are immediately displayed upon accessing a stock's page, thus requiring users to scroll down to load additional content. This posed an issue when attempting to crawl news articles programmatically, also from the stock's webpage.

To address this limitation, an alternative approach was adopted which utilized the Yahoo Finance RSS feed. RSS (Really Simple Syndication) is a publishing format for frequently updated content, allowing users to receive updates whenever new content is published [53].

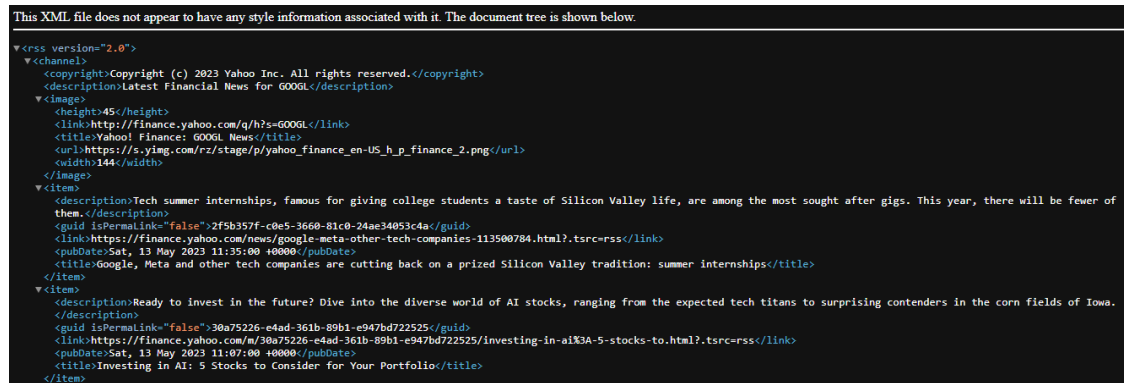


Figure 3.9: RSS Feed

First, the crawler initiates the process by calling the generic RSS Index, which provides a comprehensive listing of various financial news articles. Afterwards, the crawler modifies the URL to retrieve news specifically related to a particular stock. Hence, by formatting the RSS feed URL with a designated stock ticker, users can have access to a multitude of financial news articles pertaining to the specified stock. Figure 3.9 illustrates the details listed, such as the article's link, title, and publish date, all of which can be conveniently extracted through text manipulation methods due to the XML format. The article link allows the financial news articles to be obtained in HTML format, which is consequently parsed to string.

Ultimately, the process utilizes specialized Extractor classes, designed to extract the essential content from specific financial news articles. This is necessary because each website employs a unique HTML structure. Hence, the custom Extractor classes utilize string processing techniques to filter out the essential content of the articles while disregarding unnecessary text like HTML syntax and advertising material.

3.3.4 Machine Learning

In order to quantify the effect of a news article on a particular stock, a hybrid machine learning algorithm was adopted. This algorithm borrows the weighting and clustering concepts from ANNs and K-Means classifiers, respectively. Since market reactions differ across different types of news and stocks [48], the algorithm must operate on a per-stock level, thus training and prediction are done individually for each stock saved within the system.

Training

During the data processing phase, two invocation options are available: individual processing for a single news item or collective processing for all news currently stored in the database. At this stage, the news article content has undergone cleaning procedures, resulting in the removal of HTML syntax and advertising material. Consequently, only the essential news content remains.

The screenshot shows the 'Investment Support Tool' interface. The 'News' tab is active, displaying a list of news items on the left and a detailed view of a selected article on the right. The selected article is titled 'Pfizer Invites Public to View and Listen to a Webcast of Pfizer Discussion on Expanding'. The form includes fields for Title, Source, Timestamp, Relevant, ID, and Content. The Content field contains a detailed press release about Pfizer's webcast.

Title	Source	Timestamp	Relevant	ID	Content
Pfizer Invites Public to View and Listen to a Webcast of Pfizer Discussion on Expanding	https://finance.yahoo.com/news/pfizer-invites-public-view-listen-140000033.html	31/May/2022 16:00:00	☒	21420	NEW YORK, May 31, 2022--(BUSINESS WIRE)--Pfizer Inc. (NYSE: PFE) invites investors and the general public to view and listen to a live webcast at 3:45 p.m. EDT on Monday, June 6, 2022, with Angela Hwang, Group President, Pfizer Biopharmaceuticals Group, and Amir Malik, Chief Business Innovation Officer, Executive Vice President, who will discuss expanding access and affordability to Pfizer's medicines and vaccines. To view and listen to the webcast, visit our web site at www.pfizer.com/investors. In addition, information on accessing and registering for the webcast will be available at www.pfizer.com/investors beginning today. The transcript of the discussion will be made available on our website at www.pfizer.com/investors within 24 hours after the program. About Pfizer: Breakthroughs That Change Patients' Lives At Pfizer, we apply science and our global resources to bring therapies to people that extend and significantly improve their lives. We strive to set the standard for quality, safety and value in the discovery, development and manufacture of health care products, including innovative medicines and vaccines. Every day, Pfizer colleagues work across developed and emerging markets to advance wellness, prevention, treatments and cures that challenge the most feared diseases of our time. Consistent with our responsibility as one of the world's premier innovative biopharmaceutical companies, we collaborate with health care providers,...

Buttons: Rem Dup, New, Save, Delete, Crawl, Process, Process All, Search, Import, Total Articles: 525

Figure 3.10: News Form

The process begins by searching the news article content for tickers, which are typically enclosed within parentheses and of a short length. Once such content is found, it is checked for the presence of a colon (":") or a period ("."), as well as verifying if all letters are in uppercase. If any of these criteria are met, the code considers the text within parentheses as a ticker symbol, and the stock crawling component utilizing the Yahoo Finance Search endpoint API is initiated. If new stock tickers are found, the latter part of the stock crawler component is invoked, which extracts data relevant to those stocks from their respective Yahoo Finance webpage and saves them. This allows the tool to continuously learn and expand its data set, enhancing its ability to identify patterns associated with different stocks.

In the subsequent phase of the training process, the aforementioned hybrid machine learning algorithm is employed. The news article content is initially stored in a 2D array, with the first dimension containing the sentences and the second dimension containing the words found within the respective sentence. NER techniques are then applied to identify the occurrences of stock names, from those currently stored in the database, within the sentences. Once detected, each word in the corresponding sentence's array is assigned a weight based on the price change experienced by the specific stock on the day following the news article's publication date, hence highlighting the importance of having up-to-date data before initiating this process.

The WordANN instance associated with the mentioned stock is then retrieved. This instance contains records of different words that have been previously detected during the training process, pertaining to the mentioned stock. If words that are not currently present in the WordANN instance are discovered, a new record is created with the previously assigned weight. Conversely, if the word found is already stored in the instance, the previously assigned weight value is added to that of the pre-existing record, and its respective divisor (the total number of a specific word's occurrences, for a particular stock) is incremented. The

changes are then saved, with the expectation that the newfound data will further enhance the mechanism's performance.

Prediction

Before the prediction is invoked, a stock and prediction date must be specified. Once invoked, the WordANN and StockPriceHistoryTimeSeries objects are retrieved.

Next, the last trading day before the prediction date is determined by subtracting one day from the prediction date, excluding weekends. While this does not include bank holidays or other days on which trading is performed, it is considered enough for research purposes. The latest entry in the stock's price history for the last trading day is then obtained, from which the percentage change in price compared to the previous trading day is calculated.

The algorithm proceeds to search for news articles between the last trading day and the prediction date. It iterates through each news article, processes its content by splitting it into a 2D array of sentences and their respective words, similar to the training phase.

It then searches for occurrences of the stock name in each sentence and calculates a predicted change based on the weights associated with the words found in the sentences. The weights are obtained from the WordANN objects that are created and refined throughout the testing phase. The predicted change is calculated as the mean of the weights from the cluster of words found from the sentences that mention the specified stock, similar to a K-Means algorithm.

After processing all news articles, the prototype calculates an average predicted change by dividing the accumulated predicted change by the total number of

words counted. It adds the predicted change to the percentage change observed in the stock price on the last trading day, assuming that price changes usually have some inertia from one trading day to the next. In stocks, inertia refers to a phenomenon where the prices of stocks don't immediately respond to new information or changes in the market [54]. Finally, the predicted stock price is calculated and added to the list of predictions, which are displayed on the UI.

Chapter 4

Analysis of Results and Discussion

4.1 Evaluation Methodology

For the purpose of this research and also due to time constraints, the proof of concept implementation was evaluated on the performance of one stock exchange, XETRA. The XETRA stock exchange is one of the leading stock exchanges in Europe, containing numerous German equities and exchange traded funds. This stock exchange was chosen due to the sufficient, but not excessive number of stocks included, as well as the particular relevance of the said stocks.

News was periodically crawled during the research period. A news density function was created in the prototype, in order to list the number of articles available for each day. This allowed a 30 day evaluation period to be selected. For this research to have any academic significance, it must be bench-marked against the results of previous research. During the literature review phase of this research, it was evident that most previous research opted to predict the stock movement [9] [8] [10] [11] [32], that is the direction that the stock price will take, without going into the magnitude of the movement. This research goes further, and attempts to predict the actual price of the stock in the future, which appears to be a much rarer choice among researches, but which the research deems will have a higher significance even in real life use. However, a previous

research by Guo [1] attempts to do just that, albeit only on 3 stocks rather than a larger number, as done in this research.

However, on closer examination it turns out that while Guo's [1] calculations were most probably based on the correct MSE formula, the formula quoted on her paper is incorrect. Furthermore, the original dataset used by the researcher was not available and attempts to contact her proved futile. Hence, the only means of verifying Guo's [1] calculations were to attempt to read values from her graphs and then work back. While not exactly correct (to be expected since reading numbers off a printed graph yields limited accuracy), the results confirm that Guo [1] used the correct MSE formula and applied generalisation to abstract the accuracy levels from the actual price values of the respective stocks. Failing to do this would mean that, for instance, a 1c change on a €1 stock would have the same value of a 1c change on a €100 stock, which is not realistic. Further research by Bharti et al. [29] followed the same pattern, however used MAPE for the evaluation process. MAPE inherently includes generalisation techniques since it calculates the percentage error rate, thus making it possible to compare performance across different stocks, calculate index-wise average performance, etc. For this reason, this research will use the results achieved by Bharti et al. [29] as a benchmark in order to assess and evaluate the performance of the technology being proposed.

The evaluation process hence includes 221 stocks from the XETRA stock exchange. For each stock, the prediction for each of the 30 days selected has been calculated (excluding non-trading days), using only price data available at the close of the previous day, and news articles published on the prediction day. The MAPE between the predicted price and the closing price of the stock on the prediction day is then calculated. Hence, the respective MAPE is calculated using the following formula:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{actualPrice - predictedPrice}{actualPrice} \right| \times 100\%$$

Figure 4.1: MAPE Formula

The average prediction error for all the stocks within the stock exchange is then calculated.

4.2 Results

Running on a Core i7 with 16GB of RAM and an SSD drive, the evaluation process (comprised of the prediction of 30 days for 221 stocks) took about an hour. This gives an average single prediction time of under a second, which is considered to be a satisfactory result when taking into account all the data processing and calculations being performed.

Figure 4.2 illustrates the results of the automated evaluation procedure, which has been specifically developed for this purpose, and is designed to list the MAPE of each day of every stock. The averages for each individual stock, as well as the entire stock exchange are also calculated. Moreover, the minimum and maximum MAPE values are recorded. For reference, the full output of the procedure is being submitted with the data used for this research.

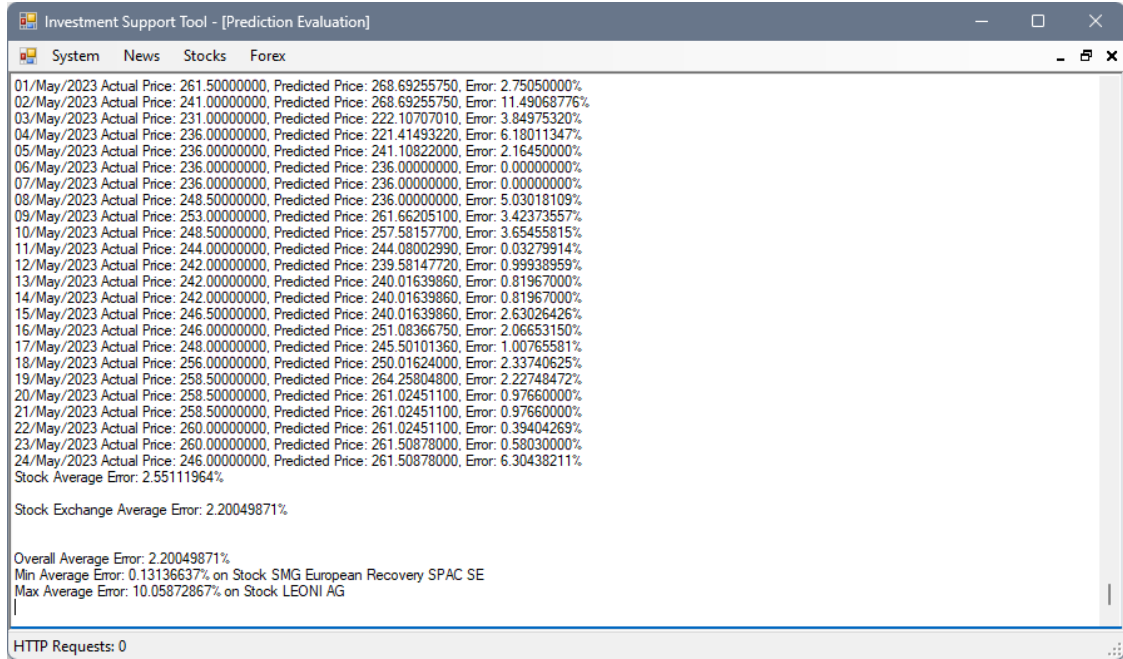


Figure 4.2: Results (MAPE)

As can be seen in Figure 4.2, the average MAPE achieved was 2.20049871%, while the best and worst MAPE values were 0.13136637% and 10.05872867% respectively. Figure 4.3 shows the distribution of the MAPE range, where it becomes evident that the vast majority (209 out of 221) are under 4%.

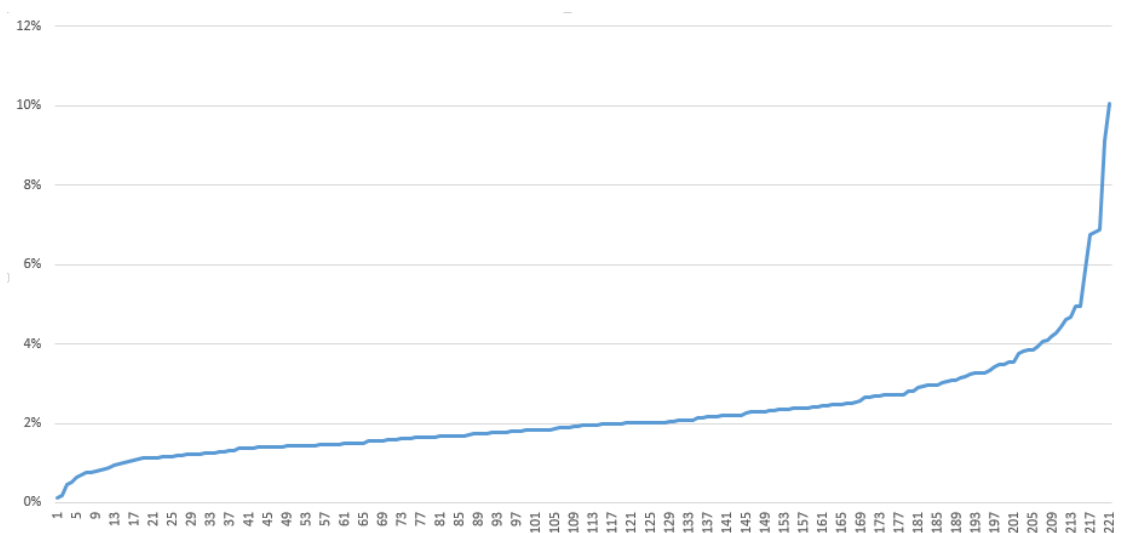


Figure 4.3: MAPE Distribution

4.3 Discussion

This section aims to discuss the results and draw academic conclusions by comparing them with those of previous research. Necessary precautions have been taken to keep the data source collection, error calculation and other parameters on level conditions as much as possible, in order to achieve a reliable comparison.

4.3.1 Data Sources

Guo [1] gathered news articles from the New York Times, and price history data from Yahoo Finance. This research utilized Yahoo Finance for both data elements, primarily to take full advantage of the APIs provided by Yahoo which are free and much more straightforward to use than scraping the New York Times website, or using its API which only allows access to news one month at a time. Having said that, Yahoo Finance's RSS News Feed offers articles from a variety of sources, including the New York Times. Guo [1] used articles from January 2015 to August 2020, while this research used data from March 2020 until May 2023.

Bharti et al. [29] used a word dataset from Cornell University to train the sentiment detection module used in their research. While the paper does not mention the sources of news and stock price data, the literature review suggests that Yahoo Finance might have been used as several of the papers cited do. Crawlers were used to extract the data from the Internet, however the details of how this was achieved (such as by HTML parsing or consumption of web services) are not given. The actual periods covered by the data used in the research are also not specified. Having said that, as long as the various parts of data involved (news, stock prices, etc.) are all from the same periods, the results are expected

to be valid. Furthermore, the paper was published quite recently in 2022, so it could be reasonably expected that the datasets cover periods around that time.

While this research uses roughly half the amount of training data used by Guo [1], it is deemed to be enough for research purposes. The model was started from a blank slate, i.e. 0 stocks, 0 stock exchanges, 0 news articles; and it was then left to crawl and process news articles, creating stocks, currencies and stock exchanges in real time as it discovers them. Neither Guo [1] nor Bharti et al. [29] mention whether their proposed technology automatically discovers stocks etc., or whether they were manually input.

4.3.2 Sample Sizes

Bharti et al. [29] did not specify the stock sample used in their research, while Guo [1] used a sample of 3 stocks, namely Amazon, Microsoft and Tesla. Bharti et al. [29] used a dataset of 10000 sentences to train the sentiment detection module. This research opted for a broader sample of 221 stocks from the XE-TRA stock exchange. This has allowed a broader evaluation of the proposed technology since some stocks might follow distinctive patterns and thus be less challenging to predict than others.

Another difference in the proposed technology and that put forward by both Bharti et al. [29] and Guo [1] is that this research proposes an increasingly more granular machine learning structure that allows article sentiment to differentiate between one stock and another. This allows different terms and words to have different meanings in conjunction with different stocks. In contrast, both Bharti et al. [29] and Guo [1] concentrated more on the extraction of pure sentiment out of the news articles themselves, without taking into account the actual stock.

4.3.3 Execution Performance

The proposed proof of concept implementation uses a custom data storage engine, that allows frequently used data to be cached, and all data to be indexed at will. Data searching is done using indexes where available, while QuickSort is used as the main sorting algorithm. This has resulted in a very good overall execution speed, which is pivotal considering the vast amount of processing that is necessary to train the model and to calculate predictions. 50,000 news articles were processed in under 7 days. Taking into account that every HTTP request is followed by a 4-5 second pause to avoid making too many requests to the Yahoo Finance servers in a short time, and that the model started from no data at all, this is considered a very good performance. During this time, 500Mb of news article data has been downloaded and analysed, while another 4.6Gb of stock price history data has also been acquired and analysed, pertaining to 19675 stocks. The resulting machine learning models totalled 404Mb.

Guo [1] explains that the actual proof of concept implementations were done in Python, relying heavily on readily available libraries. However Guo [1] does not state the actual time taken to train the model or to predict the future stock values. For this reason, no comparison could be made in terms of execution speed.

In similar fashion, Bharti et al. [29] explain that they used VADER for the sentiment analysis, and LSTM for the prediction of stock prices; but no details are given as to how it performed or the time taken to train the model and calculate predictions.

4.3.4 Prediction Performance

The following table summarizes the results achieved by this research and those achieved by Bharti et al. [29] and Guo [1]:

	Best	Average	Worst	Formula
Guo's Research	0.00109	0.013773	0.03526	MSE
Bharti et al.'s Research		4.164%		MAPE
This Research	0.131%	2.200%	10.059%	MAPE

Table 4.1: Comparison of Results

Table 4.1 summarises the results achieved by this research, Guo [1], and Bharti et al. [29]. A more detailed examination of Guo's [1] research revealed that the researcher mistakenly quoted the wrong MSE formula in her paper, thus making a direct comparison somewhat difficult. Attempts to perform reversed calculations by reading the data off the graphs illustrated in her research hinted that Guo [1] used the correct formula (as opposed to the incorrect one quoted in the paper) when doing her calculations, and also applied smoothing mechanisms as detailed in the paper itself. However, in order to ensure a more direct comparison with a similar research, Bharti et al.'s [29] research has been selected as the benchmark for prediction accuracy comparison purposes. Hence, the prototype being proposed uses the same MAPE formula used by Bharti et al. [29].

The table shows that this research achieved a significantly better average MAPE of 2.2% when compared to the 4.164% reported by Bharti et al. [29]. This shows that the approach taken by this research yields a better prediction accuracy. This could be attributed to the way the machine learning components were custom built to maximise the granularity, by practically allocating a separate structure to each stock being processed. In comparison, Bharti et al. [29] used a single machine learning model (VADER and LSTM) that is then shared by all stocks. Furthermore, the NLP and machine learning components proposed by this research

are both custom built, and designed to complement each other, thus maximising the level of collaboration and effectiveness they can achieve. Conversely, Bharti et al. [29] used separate components which were then interfaced together.

Another major difference is that Bharti et al. [29] extract sentiment out of the news article title, while this research splits the article down into sentences, and then identifies which stocks are referred to in each sentence before proceeding to process that sentence for model training. This concept enables the NLP component to be increasingly more selective, and thus avoids generalising sentiments or events that might not involve all stocks. In the next stages, sentences are broken down into words and each word is then processed separately by the custom-designed machine learning component. This approach allows the model to learn how the same word might have a different meaning when used in conjunction with different stocks, and might thus indicate a different sentiment and hence a different prediction influence. This level of detail could be one of the major advantages of the technology proposed by this research, compounded by the fact that as shown in Graph 4.3, the distribution of MAPE is not linear, and there are far less stocks with a higher MAPE than a lower one. This is also reflected in the average MAPE scores in Table 4.1 where it can be seen that while Guo's [1] MSE distribution is roughly linear, and Bharti et al. [29] did not specify the minimum and maximum MAPE achieved, the MAPE for this research is exponential.

4.3.5 Outcome

From the above results and discussion, it is evident that the technology proposed by this research improves upon the accuracy achieved by previous researchers. This could be attributed to the custom designed NLP and machine learning algorithms, which allow the proposed technology to achieve more detailed model training and hence better prediction accuracy.

Chapter 5

Conclusions and Recommendations

5.1 Conclusion

This research has concluded that while a considerable amount of previous research exists in the area of stock prediction, most of it simply targets price movements (i.e. whether the price will go up or down) rather than attempting to predict the actual price. Prediction accuracies vary, but most indicate that there is room for improvement. This research posed the following research questions:

- What level of accuracy is achievable when machine learning is applied to technical and fundamental strategies in predicting future stock prices?

This research has shown that MAPE levels of 0.131% to 10.059% are achievable using the technology being proposed. It is also probable that better results could be achieved if the model is trained with more data and over a longer period of time.

- Does custom NLP-ML integrated technology perform better than off the shelf components in stock price prediction?

Taking the custom components proposed by this research and the readily available

components used by Bharti et al. [29], the balance tips towards acknowledging that custom machine learning and NLP components can effectively increase the prediction performance.

5.2 Limitations and Recommendations

Being a proof of concept tool, the proposed technology was developed purely for research purposes. Hence, several features can be improved, built upon and even added, to further enhance its performance. It is anticipated that the proposed technology, which is currently limited to stocks, has the potential to be applied to other investment products such as Forex, Bonds, and Cryptocurrency. By expanding the prototype to include these investment products, the effectiveness of the technology across different markets within the financial domain can be assessed, whilst providing investors with additional opportunities.

Due to time constraints, the data used in the research was limited to a three-year period. However, the trends emerging from the results indicate that the accuracy of predicted prices could be further improved by evaluating the technology with larger datasets. Different time periods can exhibit different levels of market volatility, investor sentiment, and economic conditions. Hence, access to a more extensive range of news article content from different periods would enable the machine learning model to capture a broader range of market dynamics and potentially enhance its predictive capabilities.

In this research, the proposed implementation was tested over a 30-day period, also due to time constraints. The selection of this period was based on the observation of a high density of news articles published each day within the available dataset. It would be beneficial to explore how the technology performs over longer testing periods to assess its robustness and reliability in different market

conditions.

The implementation of an automated crawling and training sequence would greatly benefit the technology's continuous learning process. This would involve the automatic collection and processing of data related to stocks, their price history, news articles, and the impact of news on stocks. By eliminating the need for manual intervention, the machine learning model can continually expand its data set, improving its understanding of market dynamics and enhancing the accuracy of predictions.

The introduction of notification features, such as email or SMS alerts, would provide investors with timely information on favorable trading conditions for specific stocks. By notifying investors when the conditions align for profitable trades, the technology can increase the likelihood of making informed investment decisions at the right time, potentially leading to higher profitability.

The inclusion of more diverse data sets, such as social media platforms and investor forums, can provide valuable input to the machine learning component. These sources can offer real-time sentiment analysis, further enhancing the technology's ability to predict stock prices accurately. Although integrating these sources may require restructuring of the prototype, the effort is expected to yield better predictions and improved performance.

As the prototype is developed purely for research purposes, it may have limitations in terms of its user interface and overall user experience. Enhancing the user interface to be more intuitive and potentially porting the prototype to a web interface can greatly improve the usability and satisfaction of users. A well-designed and user-friendly interface can facilitate the adoption of the technology and enhance its practicality in real-life trading scenarios.

Modifying the proof of concept tool to allow predictions for an extended period

would be beneficial in real-life trading settings. This modification should consider stock market inertia, as the effect of news tend to diminish exponentially as the time horizon increases. Assessing the technology's performance over longer prediction periods can provide insights into its reliability and suitability for practical trading applications.

Before deploying such a technology for real-life use, obtaining feedback from professional investors who are experienced in the trading community can be valuable. Their input can help evaluate the usability and suitability of the technology in practical trading scenarios. Additionally, professional investors may provide suggestions for further improvement, considering their expertise and insights gained from their trading experience.

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